



SUPPLY CHAIN VISIBILITY & OPTIMIZATION

1st Milestone Report

Submitted by

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Supply Chain Management (SCM) is the process of planning, coordinating, and managing the flow of goods, information, and services from suppliers to end customers. It involves several interconnected activities such as procurement, manufacturing, inventory management, warehousing, transportation, distribution, and customer delivery. An efficient supply chain enables organizations to reduce operational costs, improve product availability, and enhance customer satisfaction.

In today's competitive business environment, organizations generate enormous amounts of supply chain data through daily operations. This data contains valuable information related to product sales, inventory levels, supplier performance, production, shipping, transportation, and customer demand. However, raw data is often incomplete, inconsistent, and difficult to analyze due to missing values, duplicate records, and formatting issues.

Data analytics plays an important role in transforming raw supply chain data into meaningful business insights. By analyzing operational data, organizations can identify inefficiencies, monitor inventory, evaluate supplier performance, optimize logistics, and make informed decisions. Modern Business Intelligence (BI) tools provide visual dashboards that simplify complex data and support strategic planning.

Python is one of the most widely used programming languages for data analytics due to its powerful libraries such as Pandas, NumPy, Matplotlib, and Seaborn. These libraries enable efficient data preprocessing, cleaning, transformation, and exploratory data analysis. Microsoft Power BI complements Python by providing interactive dashboards, reports, and visualizations that allow users to monitor business performance in real time.

This project, "Supply Chain Visibility & Optimization Using Python and Power BI," focuses on analyzing a real-world supply chain dataset. The project begins with preprocessing the raw data to improve its quality and consistency. After preprocessing, exploratory data analysis is performed to identify important trends related to revenue generation, inventory management, supplier performance, logistics, transportation, manufacturing, and product sales.

Finally, an interactive Power BI dashboard is developed to visualize key performance indicators (KPIs), business metrics, and operational insights. The dashboard enables organizations to improve supply chain visibility, identify operational bottlenecks, optimize inventory planning, evaluate supplier performance, and support data-driven decision-making.

CHAPTER 2

PROBLEM STATEMENT

Supply chain organizations manage large volumes of data related to products, inventory, suppliers, manufacturing, transportation, warehouses, and customer orders. In many organizations, this information is distributed across different systems, making it difficult to obtain a complete view of supply chain operations.

Poor data quality, including missing values, duplicate records, inconsistent formats, and inaccurate information, further complicates the analysis process. As a result, organizations often experience inventory shortages, excess stock, delayed deliveries, increased transportation costs, inefficient supplier management, and reduced customer satisfaction.

Traditional reporting methods provide static reports that do not support interactive analysis or real-time decision-making. Managers require a centralized platform that enables them to monitor key performance indicators, identify business trends, compare supplier performance, and analyze logistics operations efficiently.

Therefore, there is a need for a data-driven solution that preprocesses supply chain data, performs business analysis, and presents meaningful insights through interactive dashboards. This project addresses these challenges by using Python for data preprocessing and Microsoft Power BI for visualization, enabling organizations to improve supply chain visibility and optimize operational performance.

CHAPTER 3

OBJECTIVES

The primary objective of this project is to improve supply chain visibility by analyzing operational data and presenting meaningful business insights through interactive dashboards.

The specific objectives are:

- To collect and analyze a real-world supply chain dataset containing information related to products, suppliers, inventory, manufacturing, logistics, and transportation.
- To preprocess the raw dataset by identifying missing values, removing duplicate records, correcting data inconsistencies, and validating data types.
- To perform exploratory data analysis (EDA) to understand business trends, revenue generation, inventory levels, shipping performance, supplier efficiency, and manufacturing operations.
- To develop an interactive dashboard using Microsoft Power BI for visualizing key performance indicators and business metrics.
- To enable dynamic filtering and analysis of supply chain data using slicers and interactive visualizations.
- To support data-driven decision-making that improves operational efficiency, inventory management, supplier evaluation, and logistics planning.

CHAPTER 5

DATASET DESCRIPTION

5.1 Introduction

The dataset used for this project is a **Supply Chain Analysis** dataset that represents the end-to-end operations of a company's supply chain. It contains information related to products, inventory, suppliers, manufacturing, logistics, transportation, customer demographics, and revenue generation. The dataset was selected because it provides comprehensive business data required for analyzing supply chain performance and developing interactive dashboards.

The dataset was obtained from **Kaggle**, a popular platform that provides real-world datasets for data analysis and machine learning projects. It consists of structured tabular data suitable for preprocessing, exploratory data analysis, and business intelligence applications.

5.2 Dataset Overview

The dataset contains approximately **1,540 records** and **24 attributes** representing different aspects of supply chain operations. Each record corresponds to a product or transaction and contains detailed information about inventory, manufacturing, transportation, suppliers, shipping, and sales.

The dataset contains both **numerical** and **categorical** features that help analyze different business processes. Numerical features include product price, revenue generated, stock levels, shipping costs, manufacturing costs, and lead times. Categorical features include product type, supplier name, transportation mode, warehouse, customer demographics, and inspection results.

The diversity of these attributes makes the dataset suitable for evaluating supply chain performance, identifying operational inefficiencies, and generating business insights.

5.3 Dataset Attributes

The important attributes used in this project are listed below.

Attribute	Description
Product Type	Category of the product (Haircare, Skincare, Cosmetics)
SKU	Unique Stock Keeping Unit identifier
Price	Selling price of the product
Availability	Number of products available

Attribute	Description
Number of Products Sold	Total quantity of products sold
Revenue Generated	Revenue earned from product sales
Customer Demographics	Customer category information
Stock Levels	Inventory available in stock
Lead Time	Time required for product delivery
Order Quantities	Quantity ordered by customers
Shipping Times	Time taken for shipment
Shipping Carriers	Carrier responsible for delivery
Shipping Costs	Transportation cost incurred
Supplier Name	Name of the product supplier
Location	Business or delivery location
Manufacturing Lead Time	Time required for manufacturing
Production Volumes	Quantity of products manufactured
Manufacturing Costs	Cost incurred during manufacturing
Inspection Results	Product quality inspection status
Defect Rates	Percentage of defective products
Transportation Modes	Air, Road, Rail, or Sea transportation
Routes	Transportation routes used
Costs	Overall operational cost
Supplier Rating	Supplier performance rating
Warehouse	Warehouse storing the products

5.4 Purpose of the Dataset

The dataset is used to analyze various aspects of supply chain operations, including inventory management, supplier performance, manufacturing efficiency, logistics, transportation, and revenue generation.

The primary purpose of using this dataset is to understand business performance through data analysis and visualization. By analyzing different supply chain attributes, organizations can identify inefficiencies, improve operational planning, reduce transportation costs, optimize inventory levels, and make informed business decisions.

5.5 Advantages of the Dataset

The selected dataset provides several advantages for business analysis and dashboard development:

- It contains real-world supply chain information collected from multiple business operations.
- It includes both numerical and categorical attributes, enabling comprehensive analysis.
- It supports inventory management, supplier evaluation, logistics analysis, manufacturing monitoring, and revenue analysis.
- It is suitable for Python-based data preprocessing and Microsoft Power BI dashboard development.
- It helps generate meaningful business insights for improving operational efficiency and decision-making.

CHAPTER 6

SOFTWARE AND TOOLS USED

The successful implementation of this project required various software applications, programming libraries, and business intelligence tools. Each tool was selected based on its capability to support data preprocessing, analysis, visualization, and reporting.

6.1 Python

Python is a high-level, open-source programming language widely used for data analytics, machine learning, and business intelligence applications. It provides powerful libraries that simplify data preprocessing, manipulation, statistical analysis, and visualization.

In this project, Python was primarily used for importing the dataset, cleaning the data, handling missing values, removing duplicate records, verifying data types, and preparing the dataset for visualization.

6.2 Visual Studio Code (VS Code)

Visual Studio Code is a lightweight and powerful source code editor developed by Microsoft. It supports multiple programming languages and provides extensions for Python development.

In this project, VS Code was used as the development environment for writing, executing, and debugging Python programs used during data preprocessing and exploratory data analysis.

6.3 Microsoft Excel

Microsoft Excel is a spreadsheet application used for organizing, viewing, and managing structured data.

In this project, Excel was used to inspect the raw dataset, verify data entries, review preprocessing results, and store the cleaned dataset before importing it into Power BI.

6.4 Pandas Library

Pandas is one of the most popular Python libraries for data manipulation and analysis.

It provides powerful data structures such as DataFrames that simplify importing datasets, cleaning data, handling missing values, filtering records, grouping data, and performing statistical analysis.

6.5 NumPy Library

NumPy is a numerical computing library used for performing mathematical and statistical operations on numerical data.

It improves computational efficiency and supports numerical calculations required during data preprocessing and analysis.

6.6 Matplotlib Library

Matplotlib is a Python visualization library used to generate graphs and charts.

Although the final dashboard was created in Power BI, Matplotlib was used during exploratory data analysis to understand trends and relationships within the dataset.

CHAPTER 7

DATA PREPROCESSING

7.1 Introduction

Data preprocessing is one of the most important stages in any data analytics project. Real-world datasets often contain missing values, duplicate records, inconsistent formatting, incorrect data types, and unnecessary spaces. These issues can negatively affect the quality of analysis and produce misleading results if not handled properly.

The objective of data preprocessing is to improve the quality, consistency, and reliability of the dataset before performing exploratory data analysis and visualization. In this project, Python was used for preprocessing the supply chain dataset. Various preprocessing techniques such as data loading, dataset exploration, handling missing values, removing duplicate records, verifying data types, cleaning column names, and exporting the processed dataset were performed.

The cleaned dataset was then used for dashboard development in Microsoft Power BI.

7.2 Importing Required Libraries

The first step involved importing the required Python libraries. These libraries provide functions for data loading, preprocessing, analysis, and visualization.

The following libraries were used:

- **Pandas** – For importing, manipulating, and cleaning the dataset.
- **NumPy** – For performing numerical operations.
- **Matplotlib** – For generating basic visualizations during analysis.

Python Code

```
import pandas as pd  
  
import numpy as np  
  
import matplotlib.pyplot as plt
```

7.3 Loading the Dataset

The supply chain dataset stored in Microsoft Excel format was imported into Python using the Pandas library.

Python Code

```
df = pd.read_excel("supply_chain_raw_1540_rows.xlsx")
```

7.4 Understanding the Dataset

Before cleaning the dataset, it is essential to understand its structure, dimensions, column names, and statistical properties.

The following operations were performed:

Display First Five Rows

df.head()

This function displays the first five rows of the dataset to verify that it has been loaded correctly.

Display Dataset Shape

df.shape

This function returns the total number of rows and columns.

Output

(1540, 30)

This indicates that the dataset contains 1540 records and 30 attributes.

Display Column Names

df.columns

This function displays the names of all columns available in the dataset.

Display Dataset Information

df.info()

The info() function provides detailed information about the dataset, including:

- Total number of records
- Column names
- Data types
- Number of non-null values
- Memory usage

This helps identify columns that require preprocessing.

Statistical Summary

df.describe()

The describe() function generates summary statistics for numerical columns such as:

- Mean

- Standard Deviation
- Minimum Value
- Maximum Value
- Quartiles

These statistics help understand the overall distribution of the data.

7.5 Handling Missing Values

Missing values reduce the reliability of analysis and may produce inaccurate insights. Therefore, every column was checked for null values.

Python Code

```
df.isnull().sum()
```

Explanation

This command counts the number of missing values present in each column.

If missing values are found, they can be replaced using:

```
df.fillna(method='ffill', inplace=True)
```

In this project, the dataset contained very few (or no significant) missing values, so minimal preprocessing was required.

7.6 Removing Duplicate Records

Duplicate records can produce incorrect business insights because the same transaction may be counted multiple times.

The following command was used to identify duplicate rows.

Python Code

```
df.duplicated().sum()
```

To remove duplicate records:

```
df = df.drop_duplicates()
```

Explanation

Removing duplicate records ensures that every observation in the dataset is unique and prevents incorrect calculations during analysis.

7.7 Verifying Data Types

Different columns should have appropriate data types before analysis.

The following command was used:

df.dtypes

The dataset contained:

- Integer columns
- Float columns
- Object (Text) columns

Correct data types improve analytical accuracy and prevent processing errors.

7.8 Cleaning Column Names

Sometimes column names contain unnecessary spaces.

The following command removes leading and trailing spaces.

df.columns = df.columns.str.strip()

This improves consistency and avoids errors while accessing column names.

7.9 Basic Data Validation

Additional validation was performed to verify the cleaned dataset.

df.head()

This confirms that preprocessing has been completed successfully and the dataset is ready for analysis.

7.10 Saving the Cleaned Dataset

After preprocessing, the cleaned dataset was exported as a new Excel file.

Python Code

df.to_excel("clean_supply_chain.xlsx", index=False)

Explanation

The cleaned dataset was saved as clean_supply_chain.xlsx, which was later imported into Microsoft Power BI for visualization and dashboard development.

7.11 Outcome of Data Preprocessing

After completing all preprocessing steps, the dataset became clean, consistent, and ready for analysis. Missing values and duplicate records were handled appropriately, column names were standardized, and data types were verified. These preprocessing activities improved the quality and reliability of the dataset.

The cleaned dataset served as the foundation for exploratory data analysis and the development of an interactive Power BI dashboard.